

Automated Maritime Oil Spill Attribution & Evidence Platform

Complete Technical Work Breakdown Dossier: System Architecture, Technology Stack Selection, Scientific Gating Rules, and Component Engineering

Role & Module Ownership: Person 4 Lead (WebGIS Dashboard, Backend Integration & Legal Dossier)	Repository: SIH26143 / person4-backend-dashboard
Runtime Status: Verified (Python 3.11+, React 18, MapLibre GL)	Branch Status: Live, Committed & Pushed to GitHub
Date of Compilation: September 2026	Verification Test Suite: All 8 Pipeline Unit Tests Passed

1. Executive Summary & Core Challenge

Commercial vessels navigating high-seas corridors occasionally discharge oily bilge water or cargo slops under darkness or cloud cover. Historically, maritime enforcement agencies faced an insurmountable lag: by the time a synthetic aperture radar (SAR) satellite detects a slick, ocean currents and wind have drifted the oil dozens of nautical miles away from its release point, while candidate vessels have dispersed across the globe. The objective of **SIH26143** is to deliver an automated, mathematically rigorous, and legally defensible attribution pipeline connecting: **P1 (SAR Satellite ML Detection) → P2 (OpenDrift Reverse Hydrodynamic Simulation) → P3 (AIS Trajectory Correlation & Anomaly Detection) → P4 (Interactive WebGIS Command Center & Evidence Dossier)**.

Person 4 Core Mandate: Design, develop, and integrate the authoritative central WebGIS platform, the FastAPI communication hub, strict inter-module data contracts, high-performance temporal backward scrubbing, and tamper-evident PDF dossier generation.

2. Full Technology Stack: What Was Used & Architectural Rationale

Every technology was purposefully selected to satisfy strict performance, precision, and security requirements:

Technology Layer	Specific Tool / Framework	Architectural Rationale: Why It Was Chosen
Frontend Framework	React 18 with TypeScript	Guarantees compile-time type safety across complex GeoJSON spatial objects, ISO temporal strings, and API payloads. React 18's concurrent scheduler ensures fluid 60fps scrubbing during high-frequency timeline animation loops without frame dropping.
Build System	Vite 5 (ES Modules)	Delivers sub-second hot module replacement (HMR) and optimized tree-shaken production bundles. Critical when rapidly iterating on dynamic WebGL shader maps and geospatial components.

Interactive WebGIS	MapLibre GL & react-map-gl	Hardware-accelerated WebGL vector and raster tile rendering. Handles thousands of concurrent OpenDrift hydrodynamic particles, irregular multi-polygon oil spill contours, and rotated ship vectors without lag. Eliminates commercial Mapbox API token costs and usage limits.
Photorealistic Basemaps	Esri World Imagery & Ocean Bathymetry	Provides authentic high-resolution satellite Earth photography and true undersea depth contours/marine topography without watermarks, giving naval operators situational awareness.
Backend API Gateway	FastAPI (Python 3.11+) + Uvicorn	Asynchronous ASGI server capable of handling concurrent file ingestion and analytical workloads. Auto-generates interactive OpenAPI/Swagger documentation for seamless cross-team integration.
Contract Validation	Pydantic v2	Enforces strict schema validation between partner modules (P1, P2, P3, P4). Guarantees that malformed coordinates, missing timestamps, or confidence anomalies fail fast with clear HTTP 422 error vectors.
Columnar Big Data	Apache Arrow & PyArrow (Parquet)	Processes multi-gigabyte historical AIS vessel trajectory streams. Columnar compression reduces disk footprint by over 80% and accelerates spatial bounding box queries by over 20x compared to raw CSV/JSON parsing.
Relational Persistence	SQLite3	Lightweight, zero-configuration ACID-compliant SQL engine storing spill detection records, simulation vectors, and investigation audit trails without heavy external database servers.
Design System	Custom Vanilla CSS3 Glassmorphism	Avoided bloated UI frameworks (e.g. Tailwind/Bootstrap) to maintain complete pixel-level layout control, hardware-accelerated backdrop blur, high-contrast dark themes, and custom pointer drag handles.

3. Scientific Gating Rules & Business Logic

A. The Strict P1 Confidence Threshold Rule (≤ 0.30 Lookalike Gate)

Marine radar imagery frequently exhibits dark features caused by natural phenomena—such as low-wind shadows, upwelling algal blooms, or grease ice—that have radar damping ratios similar to mineral oil. **To prevent costly false investigations and judicial embarrassment, our pipeline implements an authoritative threshold gate:**

- Confidence ≤ 0.30 (Lookalike Rejection):**

 - Classified as *natural lookalike / biogenic film*, NOT mineral oil.
 - Backend halts transmission: file is **NOT forwarded to Module P2 (OpenDrift)**.
 - Endpoints PUT /api/v1/spills/{id}/drift and POST /api/v1/spills/{id}/ais/analyze return **HTTP 422 Unprocessable Entity**.
 - Dashboard displays an amber/red warning banner and completely blocks loading into the evidence map.

Confidence > 0.30 (Confirmed Mineral Oil):

 - Confirmed crude petroleum candidate.
 - Automatically forwarded to Module P2 for 12-hour reverse hydrodynamic drift modeling.

B. Strict P3 Top 3 Correlated Vessel Isolation

Module P3's ML correlation model specifically computes trajectories and association scores for the **Top 3 suspects**. In compliance with naval operational requirements, our dashboard strictly filters out all non-correlated background vessel clutter: only the 3 ranked vessels (**Rank #1 Red #ff3030: NEW YORK**, **Rank #2 Yellow #ffd21f: WANDERER**, **Rank #3 Green #28c76f: ELIZABETH ANN**) have backward trajectory trails and heading-rotated silhouettes rendered on the map.

C. Dual-Mode Timestamp Provenance (Chain of Custody)

A legal dossier is inadmissible in court if the acquisition timestamp is fabricated. Our ingestion system enforces strict provenance tagging: **satellite_metadata** (automatically extracted from GeoTIFF headers/Sentinel manifest) vs. **user_provided** (explicitly entered by an authorized maritime investigator when image metadata has been stripped). The system never invents timestamps from system clock or file creation dates.

4. Detailed Frontend Component Engineering

The dashboard is decomposed into modular, isolated, high-readability components:

Component File	Primary Functionality	Key Technical Highlights
MaritimeMap.tsx	Central WebGIS Command Map	MapLibre GL layers for detected oil polygon, 500-particle OpenDrift cloud, curved hydrodynamic streamlines, and Top 3 vessel trails with dynamic color-coding by rank.
TemporalDragger.tsx	Interactive Backward Timeline Scrubbing	Backward time scrubbing from T=0 (detection) back to T-12h. Uses HTML5 Pointer Capture API and e.stopPropagation() to enable continuous drag without map pan interference.
InvestigationPanel.tsx	Attribute & Telemetry Sidebar	Displays live interpolated proximity at T, vessel telemetry (speed, course, heading, distance to source), 30/40/30 association score breakdown, and collapsible methodology disclaimers.
SatelliteUploadModal.tsx	Module P1 Image Ingestion & Gating	Glassmorphic modal with file dropzone, interactive test scenarios (#A Confirmed 91% vs #B Lookalike 22%), timestamp provenance picker, spec grid, and collapsible P1 JSON contract inspector.
DraggablePanel.tsx	Dynamic Resizable Drawer Container	Allows operators to resize the investigation drawer horizontally between 330px and 650px with automatic 100% flex width expansion and zero black void margins.
KpiStrip.tsx & StatusBar.tsx	Real-Time KPI strip & GIS Status	Purely reactive metrics displaying slick area, detection confidence, source confidence, candidate counts, and dynamic centroid coordinates updating with every image ingestion.

5. Critical Engineering Challenges & Solutions

Bug 1: MapLibre Drag-Pan Interference on Timeline Dragger

Symptom: Clicking on the timeline slider jumped to the timestamp, but attempting to drag the thumb manually caused the MapLibre map underneath to pan, dragging the oil spill and vessels with the pointer while the slider thumb stayed frozen.

Root Cause: In HTML5 WebGL overlays, mousedown events on the slider bubbled up to the map container. MapLibre's DragPanHandler called e.preventDefault(), which cancelled the native slider's drag tracking and initiated map panning.

Solution: Added e.stopPropagation() and e.preventDefault() across onMouseDown, onPointerDown, and onTouchStart. Replaced the native slider thumb with an interactive HTML5 **Pointer Capture API** (setPointerCapture) implementation, allowing continuous 60fps dragging even if the mouse cursor drifts outside the bar.

Bug 2: Resizable Panel Width Void on Drag Expansion

Symptom: Dragging the panel left to expand it beyond 300px caused an empty black gap to appear on the right side of the cards.

Root Cause: The inner `<aside className="investigation-panel">` had a legacy hardcoded width: `var(--panel-w) (300px)` CSS rule that did not scale with the parent container.

Solution: Updated `.investigation-panel` and `.panel` in `global.css` to `width: 100%, flex: 1, and min-width: 0`. All spec grids and telemetry cards now stretch dynamically across any dragged width (330px–650px).

6. Quality Assurance, Testing & Git Deployment

The codebase has undergone full automated validation and is committed to the remote repository:

Verification Test	Command Executed	Status & Output
Frontend TypeScript Compilation	<code>npm run build</code>	PASSED: 0 TypeScript errors, 87 modules transformed, production bundle built cleanly in 2.4s.
Confidence Rule Unit Tests	<code>python tests/test_confidence_rule.py</code>	PASSED: All 8 backend validation tests passed (lookalike rejection, HTTP 422 gating, P2 forwarding).
Remote Git Synchronization	<code>git push origin person4-backend-dashboard</code>	LIVE ON GITHUB: Commit 239633e & fd48be8 successfully pushed. Working tree 100% clean.

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